



AI SPI Inference API

IPC

Conda uv Python venv Windows

	ai-spi-inference-api 0.1.0
	NVIDIA AIPC
Python	>= 3.12, < 3.13
	2026-08-31
	uv + uv Conda Pack

README pyproject.toml uv.lock Windows

	4	A uv	AIPC	Python uv	7	Conda Pack
.venv						

- 1-3
- 4-7 uv venv Conda
- 8-11 AIPC TensorRT Windows
- 12-13
- 14 troubleshooting

- AIPC USB
- AIPC Windows 10/11 Windows Server x86-64 NVIDIA GPU Linux NSSM
- SPI LAN/SMB LAN watch_root
- Git .pt .onnx .engine

Python venv	TensorRT engine	TensorRT	GPU
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1.

8000 8002 durable pipeline Ingest Inference Publisher AIPC SQLite WAL
API 5050

CC	CCC	CC	CCCCC
Publisher	-	Primary 23	
PatchCore anomaly	8000	anomaly_score	required
Distance detection	8002	min_pad_distance	required
Inference	-	manifest	
Ingest	-	READY	
Paste detection	8001	paste_pixels	
Compatibility API	5050	/process /health /ready	

□□□□Publisher -> 8000/8002 health=healthy -> Inference -> Ingest Publisher
deadline fallback

2.

CC	CCCC	CCCCC
Python	>=3.12,<3.13 3.12	Python 3.12 x64 Conda
PyTorch	2.7.1 + CUDA 12.8 index	torch 2.7.1+cu128 torchvision 0.22.1+cu128
ONNX Runtime	onnxruntime-gpu 1.19.0	wheel CUDA/cuDNN
PyCUDA	2024.1	wheel
TensorRT	10.x 10.1.x	driver/CUDA NVIDIA Python wheel
uv	pyproject.toml + uv.lock	uv.exe uv.lock
Windows runtime	wheel	NVIDIA driver VC++ Redistributable VS Build Tools

uv.lock	uv.lock	requires-python=3.12	uv.lock	.gitignore	.venv	Python 3.10
uv.lock	AIPC	Python 3.12	.venv			

2.1 AIPC

- CPU x86-64 GPU compute capability
- NVIDIA driver CUDA runtime/toolkit cuDNN TensorRT Python binding
- AIPC D:\Dre\JQ_SPI_02_AI_API
- SPI watch_root machine return state/result/log
- SHA256 input shape batch profile
-

3.

CC	CC	CC	CCCCC
uv	uv.lock pyproject.toml	uv	cache
venv + wheelhouse	Python/pip wheel	PyCUDA/TensorRT	
Conda + conda-pack	Python	OS/	AIPC Python/uv

uv	Conda Pack	wheelhouse	AIPC	OS	Python minor	GPU
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4. A uv

4.1 Windows

- 1. AIPC Python 3.12 x64 NVIDIA driver/CUDA/TensorRT
- 2. uv.exe

```
cd /d D:\build\ai_spi_inference_api
set UV_CACHE_DIR=D:\offline_bundle\uv-cache
uv --version
uv lock --check
uv venv --python 3.12 --clear
uv sync --locked --extra cuda --no-group dev
```

- 3. AIPC tensorrt-export dev paste Git

```
uv sync --locked --extra cuda --extra tensorrt-export --no-group dev
rem uv sync --locked --extra cuda --group dev
```

- 4. GPU provider

```
uv run python scripts\check_env.py
uv run python -c "import torch; print(torch.__version__, torch.version.cuda, torch.cuda.is_available())"
uv run python -c "import onnxruntime as ort; print(ort.get_available_providers())"
uv run pytest tests\unit tests\integration
```

uv --offline	.venv	UV_CACHE_DIR	4.3
uv sync			

4.2

```
AI_SPI_OFFLINE_2026-08-31\
project\ uv.lock
uv\uv.exe
uv-cache\ uv cache
installers\ Python 3.12 driver VC++ CUDA TensorRT NSSM
wheelhouse\ wheels CUDA torch
models\ engine.json SHA256SUMS.txt
config\ai_server.aipc.json
MANIFEST.txt
SHA256SUMS.txt
```

- .venv log SQLite state pipeline_results config/ai_server.json
- uv.lock .gitignore git clone/zip tracked files
- SHA256 AIPC

4.3 AIPC uv

- 5. AIPC SHA256 driver VC++ Python 3.12 CUDA/TensorRT
- 6. project uv.exe uv-cache USB SMB

```
cd /d D:\Dre\JQ_SPI_02_AI_API
```

```
set UV_CACHE_DIR=D:\Dre\AI_SPI_OFFLINE\uv-cache
set UV_OFFLINE=1
D:\Dre\AI_SPI_OFFLINE\uv\uv.exe venv --python C:\Python312\python.exe --clear
D:\Dre\AI_SPI_OFFLINE\uv\uv.exe sync --frozen --offline --extra cuda --no-group dev --no-python-downloads
```

AIPC TensorRT --extra tensorrt-export .engine TensorRT
runtime/Python binding onnx/onnxslim

7. Python PATH

```
set PYTHON_EXE=D:\Dre\JQ_SPI_02_AI_API\.venv\Scripts\python.exe
%PYTHON_EXE% --version
%PYTHON_EXE% scripts\check_env.py
```

5. B venv + wheelhouse

AIPC Python 3.12 .venv wheelhouse Python
venv .venv

5.1 wheelhouse

8. Windows/Python/GPU

```
py -3.12 -m venv D:\build\spi-wheel-build
D:\build\spi-wheel-build\Scripts\activate.bat
python -m pip install --upgrade pip wheel
python -m pip install torch==2.7.1 torchvision==0.22.1 --index-url https://download.pytorch.org/whl/cu128
python -m pip install -r requirements-cuda.txt
python -m pip freeze > offline_bundle\requirements-resolved-win-py312.txt
```

9. / wheel PyCUDA wheel CUDA headers MSVC Build Tools
sdist AIPC

```
python -m pip download -d offline_bundle\wheelhouse ^
-r offline_bundle\requirements-resolved-win-py312.txt ^
--extra-index-url https://download.pytorch.org/whl/cu128
python -m pip wheel -w offline_bundle\wheelhouse pycuda==2024.1
```

10. TensorRT /whl requirements-cuda.txt AIPC NVIDIA

PyPI index AIPC --no-index

5.2 AIPC

```
cd /d D:\Dre\JQ_SPI_02_AI_API
py -3.12 -m venv .venv
.venv\Scripts\python.exe -m pip install --no-index ^
--find-links D:\Dre\AI_SPI_OFFLINE\wheelhouse ^
-r D:\Dre\AI_SPI_OFFLINE\requirements-resolved-win-py312.txt
.venv\Scripts\python.exe -m pip check
.venv\Scripts\python.exe scripts\check_env.py
```

PowerShell Activate.ps1 .venv\Scripts\python.exe
PYTHON_EXE

6. C Conda

setup.bat py312_cu128_j15 PyTorch CUDA requirements.txt
AIPC

```
cd /d D:\build\ai_spi_inference_api
setup.bat
conda run -n py312_cu128_j15 python scripts\check_env.py
conda run -n py312_cu128_j15 python -m pip check
```

requirements.txt	Conda/plain-pip	legacy mirror	pyproject.toml	uv.lock
Conda	uv			

7. Conda Pack Python AIPC

7.1

```
conda install -n base -c conda-forge conda-pack -y
conda pack -n py312_cu128_j15 ^
-o D:\offline_bundle\py312_cu128_j15.zip ^
--format zip
```

- OS Windows Linux
- torch/CUDA ONNX provider TensorRT import
- Conda Pack NVIDIA driver

7.2 AIPC

```
powershell -NoProfile -Command ^
"Expand-Archive -Path D:\AI_SPI_OFFLINE\py312_cu128_j15.zip -DestinationPath D:\Dre\envs\py312_cu128_j15"
D:\Dre\envs\py312_cu128_j15\Scripts\activate.bat
D:\Dre\envs\py312_cu128_j15\Scripts\conda-unpack.exe
set PYTHON_EXE=D:\Dre\envs\py312_cu128_j15\python.exe
%PYTHON_EXE% scripts\check_env.py
```

conda-unpack	zip
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8.

8.1

CC	CC	CC/CCCC
	app scripts tests *.py *.bat README pyproject.toml uv.lock	.git cache .venv
	ai_server.example.json AIPC config	
	PatchCore center pad paste engine.json	artifact
	state/log/result	SQLite WAL job/result

CC	CC	CC/CCCC
	driver VC++ Python/Conda CUDA TensorRT uv NSSM	AIPC

8.2 SHA256

```
rem PowerShell
Get-ChildItem D:\offline_bundle -Recurse -File |
Get-FileHash -Algorithm SHA256 |
ForEach-Object { "$($_.Hash) *$($_.Path)" } |
Set-Content D:\offline_bundle\SHA256SUMS.txt
```

AIPC MANIFEST Git commit uv Python torch/CUDA onnxruntime
providers TensorRT driver GPU SHA256 SHA256

TensorRT engine Python pickle .pt/.pth/.ckpt PatchCore .pt
--trust-pickle

9. AIPC

9.1 AIPC

- 11. config\ai_server.example.json D:\Dre\config\ai_server.aipc01.json
- 12. AI_CONFIG_PATH /worker/ API process

```
set AI_CONFIG_PATH=D:\Dre\config\ai_server.aipc01.json
set PYTHON_EXE=D:\Dre\JQ_SPI_02_AI_API\venv\Scripts\python.exe
```

□□□□	□□□□	□□□□
paths.external_output_root	Primary	NSSM
paths.backup_output_root	AIPC	SMB SLA/
pipeline.watch_root	SPI timestamp durable Ingest	legacy watch_root
pipeline.database_path	AIPC NTFS worker	UNC/SMB WAL
pipeline.staging_root	null raw backup	30
pipeline.result_root	AIPC manifest/processed	
model_clients	8000/8002 enabled+required URL/	required service 23
output	durable is_pass_only + machine_return	
reliability	deadline 30 reserve 5 lease	reserve/lease
logging.log_dir	stage system.*	

9.2

```
%PYTHON_EXE% -c "from pathlib import Path; from app.core.config import load_config; c=load_config(Path(r'%AI_CONFIG_PATH%'));
print(c.model_dump_json(indent=2))"
%PYTHON_EXE% -m app.pipeline.publisher --once --config "%AI_CONFIG_PATH%"
```

Publisher preflight --once

9.3

```
D:\Dre\JQ_SPI_02_AI_API\      .venv
D:\Dre\config\
D:\Dre\JQ_SPI_02_AI_API\state\ SQLite + WAL
D:\Dre\JQ_SPI_02_AI_API\pipeline_results\
D:\Dre\JQ_SPI_02_AI_API\log\
D:\Dre\JQ_SPI_02_AI_API\backup\
```

- state/result/log/backup machine return / /
- external_output_root/watch_root SMB UNC Windows service
- SQLite database_path UNC/SMB AIPC

10. TensorRT

10.1

CC	CC	AIPC CCCC
.pt	PyTorch / Ultralytics	CUDA torch PatchCore pickle --trust-pickle
.onnx	ONNX Runtime GPU	available providers CUDAEExecutionProvider YOLO batch 8
.engine	TensorRT/PyCUDA	AIPC

10.2

```
set SPI_MODEL_DEVICE=cuda:0
set PATCHCORE_MODEL_PATH=models\patchcore\model_fp16.engine
set DISTANCE_CENTER_MODEL_PATH=models\distance\center.engine
set DISTANCE_PAD_MODEL_PATH=models\distance\pad.engine
set PATCHCORE_MODEL_ARGS=
start_model_services.bat
```

NSSM AppDirectory PatchCore .pt
 PATCHCORE_MODEL_ARGS=--trust-pickle ...

10.3 AIPC TensorRT

TensorRT plan TensorRT GPU NVIDIA AIPC engine
 TensorRT 10.1.x artifact profile batch engine.json

```
%PYTHON_EXE% -m pip install --no-index --find-links D:\AI_SPI_OFFLINE\wheelhouse ^
-r requirements-tensorrt-export.txt
```

```
convert_to_tensorrt.bat yolo ^
--input models\distance\pad.pt ^
--output models\distance\pad.engine ^
--task detect --precision fp16 --batch 8 --imgsz 640 640 ^
--workspace-gib 4 --save-onnx models\distance\pad.onnx
```

- center pad YOLO service batch 8 ONNX 1/ /8 profile 8
- PatchCore NCHW RGB float32 [0,1] input size center crop ImageNet/custom normalization score output
- SPI PyTorch/ONNX/TensorRT 22/23 engine

11.

11.1

13. PYTHON_EXE AI_CONFIG_PATH 03 Publisher
14. start_model_services.bat 8000 8002 HTTP 200 JSON status healthy
backend/model/device
15. 02 Inference 01 Ingest

```
start "SPI Publisher" 03_pipeline_publisher.bat
start_model_services.bat
powershell -NoProfile -Command "Invoke-RestMethod http://127.0.0.1:8000/health | ConvertTo-Json -Depth 6"
powershell -NoProfile -Command "Invoke-RestMethod http://127.0.0.1:8002/health | ConvertTo-Json -Depth 6"
start "SPI Inference" 02_pipeline_inference.bat
start "SPI Ingest" 01_pipeline_ingest.bat
```

	Durable mode	app.pipeline worker	Compatibility mode	scan_jobs.py + app.main	legacy
scanner	Stage 01	share			

11.2 NSSM

worker model path/device AppParameters

AppEnvironmentExtra

```
set REPO=D:\Dre\JQ_SPI_02_AI_API
set PYTHON_EXE=%REPO%\venv\Scripts\python.exe
set CONFIG=D:\Dre\config\ai_server.aipc01.json

nssm install SPI_03_Publisher "%PYTHON_EXE%" -m app.pipeline publisher
nssm set SPI_03_Publisher AppDirectory "%REPO%"
nssm set SPI_03_Publisher AppEnvironmentExtra "AI_CONFIG_PATH=%CONFIG%"
nssm set SPI_03_Publisher Start SERVICE_AUTO_START

nssm install SPI_02_Inference "%PYTHON_EXE%" -m app.pipeline inference
nssm set SPI_02_Inference AppDirectory "%REPO%"
nssm set SPI_02_Inference AppEnvironmentExtra "AI_CONFIG_PATH=%CONFIG%"

nssm install SPI_01_Ingest "%PYTHON_EXE%" -m app.pipeline ingest
nssm set SPI_01_Ingest AppDirectory "%REPO%"
nssm set SPI_01_Ingest AppEnvironmentExtra "AI_CONFIG_PATH=%CONFIG%"
```

- watch/output/backup/state/result/log UNC
- stdout/stderr system.ingest system.inference system.publisher
- Publisher Inference/Ingest delayed start

12.

12.1 GPU

- [] %PYTHON_EXE% --version 3.12.x sys.executable
- [] scripts/check_env.py required base package OK
- [] torch.__version__ 2.7.1 CUDA build torch.cuda.is_available() True
- [] onnxruntime providers CUDAExecutionProvider ONNX GPU
- [] tensorrt pycuda import engine driver/CUDA/TensorRT
- [] pip check uv sync --locked --offline

12.2

- [] AI_CONFIG_PATH AIPC Pydantic load_config
- [] SQLite backup result log AIPC watch/output
- [] 8000/8002 health status=healthy device
- [] Stage 03 Stage 02 Stage 01 legacy scanner
- [] SHA256 MANIFEST

12.3

16. config watch_root output_root SPI share
17. 14 timestamp job CSV OpenCV
18. SQLite READY / INFERENCING / RESULT_READY / PRIMARY_RETURNED / DONE
19. Primary external_output_root is_pass delimiter quoting line ending
20. backup immutable raw ai_result/returned processed manifest.json log latency
deadline_met
21. required model reason required_model_failure/timeout
publish_reserve_reached 23

fail-safe	AI	required model	23
-----------	----	----------------	----

13.

13.1

- 22. uv.lock config SHA256
- 23. scanner/ API share
- 24. config SQLite
- 25. Publisher -> models -> Inference -> Ingest job deadline

13.2

- config/ /
- .engine TensorRT/GPU engine
- worker SQLite/WAL
- Primary job at-least-once CSV
- manifest stage logs SQLite job state audit event ledger

14. Troubleshooting

Python /GPU/provider stage/model logs

```
where python
echo %PYTHON_EXE%
%PYTHON_EXE% --version
%PYTHON_EXE% scripts\check_env.py
nvidia-smi
```

14.1 Python 3.12 is required

- resolve_python.bat .venv uv requires-python
- 3.12 .venv Python 3.10
- Python 3.12 set PYTHON_EXE .venv AIPC

14.2 uv --offline

- cache miss No solution found failed to fetch
- uv-cache platform marker wheel uv.lock uv Python/OS/ paste Git
- Windows x86-64 UV_CACHE_DIR uv.exe uv.lock paste
- Git commit / wheel

14.3 pip --no-index No matching distribution

```

□□□ sdist wheel tag cp312/win_amd64
□□□ wheelhouse macOS/Linux Python PyCUDA/TensorRT wheel
□□□ Windows Python 3.12 x64 / wheel pip debug --verbose tag PyCUDA CUDA
headers/MSVC TensorRT wheel NVIDIA

```

14.4 DLL load failed /

```

□□□ import torch onnxruntime tensorrt pycuda cv2
□□□ VC++ runtime CUDA/cuDNN/TensorRT DLL PATH 32/64
□□□ nvidia-smi where DLL PATH DLL

```

14.5 torch.cuda.is_available() = False

```

□□□ torch import cuda:0
□□□ CPU wheel driver GPU
□□□ torch.__version__ +cu128 torch.version.cuda nvidia-smi cu128 wheel

```

14.6 ONNX Runtime CUDAAExecutionProvider

```

□□□ available providers CPUExecutionProvider cuda:N CPU
□□□ onnxruntime onnxruntime-gpu CUDA/cuDNN DLL
□□□ CPU/GPU onnxruntime-gpu==1.19.0 provider native DLL CPU fallback

```

14.7 TensorRT engine

```

□□□ magic tag compute capability binding/profile engine validity
□□□ engine OS/GPU/TensorRT runtime 10.1.x
□□□ engine.json SHA256 AIPC TensorRT 10.x 10.1.x batch/profile preprocessing

```

14.8 /health 200

```

□□□ HTTP JSON status initializing/unhealthy
□□□ TensorRT engine GPU OOM backend
□□□ payload status=healthy /stderr model path device free VRAM Stage 02/01

```

14.9 Worker / Config invalid

```

□□□ pipeline.enabled preflight Pydantic validation error

```

□□□watch_root required model duplicate name/target_column deadline/reserve/lease output
durable contract

□□□ 9.2 load_config durable is_pass_only + machine_return worker_lease > deadline
publisher heartbeat < lease < reserve

14.10 Ingest READY

□□□timestamp folder READY

□□□ 14 settle CSV / / key

□□□ system.ingest component_name Array_id Pad_id image_name_template --once --date
YYYY-MM-DD

14.11 23

□□□API 200 fail-safe 23

□□□required model failure/timeout key / publish reserve

□□□ backup/.../ai_result/manifest.json reason/errors/timing system.inference logs
optional

14.12 SQLite locked /

□□□ WAL worker queue

□□□database_path SMB/UNC worker config/database /

□□□ AIPC NTFS AI_CONFIG_PATH worker

14.13 Job PUBLISHING PRIMARY_RETURNED

□□□ DONE

□□□PUBLISHING machine return/SMB PRIMARY_RETURNED Primary ai_result
finalize

□□□ system.publisher share Publisher lease inference Primary

14.14 NSSM

□□□ FileNotFound/Permission denied

□□□AppDirectory AppEnvironmentExtra

□□□ AppDirectory PYTHON_EXE/AI_CONFIG_PATH/model path SMB UNC

14.15 Primary 30

□□□deadline_met=false share timeout Publisher

□□□SMB GPU/CPU staging_root publish cutoff
 □□□ share IO staging_root null ready_at stage latency 30 soft real-time AIPC/SMB

14.16 high cover

□□□ high cover
 □□□paste 8001 cover% paste_pixels Width/Length pad_area
 □□□ MobileSAM/paste extra 8001 enabled=true 30

15.

15.1

```
%PYTHON_EXE% --version
%PYTHON_EXE% -c "import sys; print(sys.executable); print(sys.version)"
%PYTHON_EXE% scripts\check_env.py
%PYTHON_EXE% -m pip check
%PYTHON_EXE% -c "import torch; print(torch.__version__, torch.version.cuda, torch.cuda.is_available())"
%PYTHON_EXE% -c "import onnxruntime as ort; print(ort.__version__, ort.get_available_providers())"
%PYTHON_EXE% -c "import tensorrt as trt; print(trt.__version__)"
```

15.2 Worker

```
01_pipeline_ingest.bat --once --date 2026-08-31
02_pipeline_inference.bat --once --worker-id manual-inference
03_pipeline_publisher.bat --once
```

--once	config	job
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15.3

```
Invoke-RestMethod http://127.0.0.1:8000/health | ConvertTo-Json -Depth 6
Invoke-RestMethod http://127.0.0.1:8002/health | ConvertTo-Json -Depth 6
Invoke-RestMethod http://127.0.0.1:5050/ready | ConvertTo-Json -Depth 6
```

5050 /ready API durable worker heartbeat/queue stage SQLite system.* logs
manifest

16.

- RE DME.md README.md
- pyproject.toml / uv.lock Python marker
- config/ai_server.example.json
- Python 3.12 venv <https://docs.python.org/3.12/library/venv.html>
- uv CLI offline / frozen / no-python-downloads <https://docs.astral.sh/uv/reference/cli/>

- **uv cache** <https://docs.astral.sh/uv/concepts/cache/>
- **pip download** https://pip.pypa.io/en/stable/cli/pip_download/
- **Conda Pack** <https://conda.github.io/conda-pack/>
- **NVIDIA TensorRT Engine Compatibility** <https://docs.nvidia.com/deeplearning/tensorrt/latest/inference-library/engine-compatibility.html>
- **NVIDIA TensorRT Support Matrix** <https://docs.nvidia.com/deeplearning/tensorrt/latest/getting-started/support-matrix.html>

- Python CUDA PyTorch ONNX Runtime TensorRT schema
- uv.lock .gitignore
- AIPC SHA256 config SHA256